

Table 2: Test set and adversarial accuracy with 100 samples during the attack and allowed perturbation strength of 1.5 on FashionMNIST and 0.3 on CIFAR10 for increasing number of samples used during prediction. We estimated the average accuracy 10 times and report the average and standard deviation.

$ Z $	FASHIONMNIST				CIFAR10	
	BNN	IM	SIN 0.05	SIN 0.1	DR 0.3	DR 0.6
TEST SET ACCURACY						
1	83.03 $\pm$ 0.12	79.15 $\pm$ 0.27	86.98 $\pm$ 0.17	85.76 $\pm$ 0.17	92.08 $\pm$ 0.13	92.68 $\pm$ 0.15
5	84.81 $\pm$ 0.14	84.29 $\pm$ 0.21	88.21 $\pm$ 0.09	87.49 $\pm$ 0.15	92.57 $\pm$ 0.05	93.51 $\pm$ 0.09
10	85.03 $\pm$ 0.13	85.02 $\pm$ 0.18	88.47 $\pm$ 0.06	87.92 $\pm$ 0.10	92.66 $\pm$ 0.06	93.60 $\pm$ 0.06
100	85.21 $\pm$ 0.06	85.72 $\pm$ 0.08	88.63 $\pm$ 0.07	88.27 $\pm$ 0.06	92.74 $\pm$ 0.03	93.69 $\pm$ 0.04
ADVERSARIAL ACCURACY						
1	35.76 $\pm$ 0.92	44.67 $\pm$ 1.19	36.15 $\pm$ 0.79	40.63 $\pm$ 0.79	42.64 $\pm$ 0.39	47.18 $\pm$ 0.70
5	36.19 $\pm$ 0.47	47.08 $\pm$ 0.73	35.92 $\pm$ 0.51	40.57 $\pm$ 0.44	42.73 $\pm$ 0.31	47.00 $\pm$ 0.38
10	36.30 $\pm$ 0.50	47.23 $\pm$ 0.62	36.06 $\pm$ 0.41	39.93 $\pm$ 0.66	42.69 $\pm$ 0.37	47.01 $\pm$ 0.33
100	36.50 $\pm$ 0.39	47.96 $\pm$ 0.24	35.91 $\pm$ 0.23	39.67 $\pm$ 0.29	42.82 $\pm$ 0.17	47.02 $\pm$ 0.20

6 Conclusion

In this work we have stressed the fact that stochastic neural networks (SNNs) (and stochastic classifiers in general) often depend on samples and thus their predictions are random variables themselves. For gradient-based adversarial attacks this means that the attack is calculated based on one realization of the stochastic network (which depends on multiple samples of the random variables used in the network) and applied to another which is used for inference. We derived a sufficient condition for this inference network to be robust against the calculated attack. This allowed us to identify the factors that lead to an increased robustness of stochastic classifiers: i) larger prediction margins ii) a smaller norm of the gradient estimates and iii) higher angles between the attack direction and the direction to the closest decision boundary during inference. The observed angles depend inverse proportionally on the norm of the expected gradient and proportionally on the variance of the gradient estimates. This variance can be reduced by increasing the sample size. These insights enable us to explain previously reported empirical findings for SNNs from a geometrical perspective, e.g. that the robustness of SNNs is higher than the robustness of their deterministic counterparts even for strong attacks that are based on several samples [He et al., 2019, Eustratiadis et al., 2021], why regularization of the gradient variance [Bender et al., 2020], norm of the mean gradient, and angle [Lee et al., 2022] improves the adversarial robustness, and last but not least why increasing the sample size during attack is important to exploit its potential [Athalye et al., 2018]. Therefore, our work poses a general applicable and simple framework, that helps understanding the mode of operation of existing stochastic defense mechanisms, even if they were motivated from a different point of view. Moreover, we derived a justification for the common practice of choosing the sample size during inference in a way that balances its prediction certainty against the computational cost. Finally, we believe our findings will be useful to evaluate and compare the robustness of different models, since they point out that they might require different amounts of samples during attack to sufficiently reduce variance and hope that they will help to improve the robustness of stochastic classifiers in future.

Potential negative societal impact and limitations. Since we do not propose a new attack strategy, but contribute to a better understanding of the robustness of SNNs and advise to cautiousness when determining sample sizes, we do not see negative impact.

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